

**SI 427 PROBLEM SET**  
**AUTUMN 2025**

**Week 1 (28 Jul – 1 Aug)**

- 1.1. Complete the proof of the inclusion–exclusion principle

$$\mathbb{P}\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n \mathbb{P}(A_k) - \sum_{1 \leq k_1 < k_2 \leq n} \mathbb{P}(A_{k_1} \cap A_{k_2}) + \cdots + (-1)^{n+1} \mathbb{P}\left(\bigcap_{k=1}^n A_k\right).$$

Note that we had discussed the base case in the lecture.

- 1.2. Complete the proof of Bonferroni's inequality

$$\sum_{i=1}^n \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) \leq \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

Note that we had discussed the base case in the lecture.

- 1.3. Prove the following generalization of Boole's inequality: If  $\{A_j\}_{j \geq 1}$  is a countable sequence of sets in the  $\sigma$ -algebra, then

$$\mathbb{P}\left(\bigcap_{j \geq 1} A_j\right) \geq 1 - \sum_{j \geq 1} \mathbb{P}(A_j^c).$$

- 1.4. If  $A, B, C$  are three events, we say that  $A$  and  $B$  imply  $C$  if  $A \cap B \subseteq C$ . Prove that

$$\mathbb{P}(A^c) + \mathbb{P}(B^c) \geq \mathbb{P}(C^c).$$

- 1.5. Let  $\mathcal{B}(\mathbb{R})$  be the Borel  $\sigma$ -algebra of  $\mathbb{R}$ , which is the smallest  $\sigma$ -algebra containing all open sets of  $\mathbb{R}$ . Let  $\mathbb{Q}$  be the set of all rational numbers. Show that  $\mathbb{Q} \in \mathcal{B}(\mathbb{R})$ .

**Week 2 (4 Aug – 8 Aug)**

- 2.1. For a sequence  $\{A_n\}_{n \geq 1}$  of sets in a  $\sigma$ -algebra such that  $A_1 \supseteq A_2 \supseteq \cdots$ , prove that  $\lim_{n \rightarrow \infty} A_n$  exists.
- 2.2. A point is chosen uniformly randomly from the interval  $[0, 1]$ , and a second point is chosen uniformly randomly on the right of the first point. What is the probability that the three segments form a triangle?
- 2.3. Find the probability that the sum of two randomly chosen positive real numbers (both  $\leq 1$ ) does not exceed 1, and their product does not exceed  $2/9$ .

- 2.4. Given a set of  $n$  distinct objects, a set of  $r \geq 2$  objects is chosen with replacement. What is the probability that the object chosen in the first pick is chosen at least twice?
- 2.5. Given a set of  $n$  distinct objects, a set of  $r \geq 2$  objects is chosen with replacement. What is the probability that the object chosen in the first pick is chosen at most twice?

### Week 3 (11 Aug – 15 Aug)

- 3.1. Ram and Shaam are among a group of  $2n$  boys. If all these boys are divided into two equal subgroups, find the probability that Ram and Shaam are in  
 (a) Different groups,  
 (b) Same group.
- 3.2. Each permutation of the digits 1, 2, 3, 4, 5, 6 determines a six-digit number. If the numbers corresponding to all possible permutations are listed in increasing order of magnitude, find the 319<sup>th</sup> number on this list.
- 3.3. In a population of  $N$  elements,  $R$  are red and  $W = N - R$  are white. A group of  $n$  elements is selected at random. Find the probability that the group so chosen will contain exactly  $r$  red elements.
- 3.4. An urn contains  $R$  red and  $W$  white marbles. Marbles are drawn from the urn one after another without replacement. Let  $A_k$  be the event that a red marble is drawn for the first time on the  $k^{\text{th}}$  draw. Show that

$$P(A_k) = \left( \frac{R}{R + W - k + 1} \right) \prod_{j=1}^{k-1} \left( 1 - \frac{R}{R + W - j + 1} \right).$$

Let  $p$  be the proportion of red marbles in the urn before the first draw. Show that  $P(A_k) \rightarrow p(1-p)^{k-1}$  as  $R + W \rightarrow \infty$ .

- 3.5. Consider a town with  $N$  people. A person sends two letters to two separate people, each of whom is asked to repeat the procedure. Thus for each letter received, two letters are sent out to separate persons chosen at random (irrespective of what happened in the past). What is the probability that in the first  $n$  stages the person who started the chain letter game will not receive a letter?

### Weeks 4 & 5 (18 Aug – 29 Aug)

- 4.1. A coin is tossed till heads appears. Write down a corresponding random variable, and its distribution function.
- 4.2. Let  $A$  and  $B$  be two events such that  $P(A) = p_1 > 0$ ,  $P(B) = p_2 > 0$  and  $p_1 + p_2 > 1$ . Show that  $P(B|A) \geq 1 - \frac{1-p_2}{p_1}$ .
- 4.3. Each of  $n$  urns contains four white and six black balls, while another urn contains five white and five black balls. An urn is chosen at random from the  $n + 1$  urns, and two balls are drawn from it (without replacement), both being black. The probability that

five white and three black balls remain in the chosen urn is  $1/7$ . Find  $n$ .

4.4. On  $\Omega = (0, 1)$ , each subinterval is assigned a probability equal to its length. Find a necessary and sufficient condition for two events to be independent.

4.5. Let  $X$  be an RV. Is  $|X|$  also an RV? If  $X$  takes only non-negative values, is  $\sqrt{X}$  an RV?

### Week 6 (1 Sep – 5 Sep)

6.1. Let  $X$  be an RV with PMF

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, 2, \dots, n, \quad 0 \leq p \leq 1.$$

Find the PMFs of

- (a)  $Y = aX + b$
- (b)  $Y = X^2$

6.2. Let  $X$  be an RV with PDF

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ \frac{1}{2} & \text{if } 0 < x \leq 1, \\ \frac{1}{2x^2} & \text{if } 1 < x < \infty. \end{cases}$$

Find the PDF of the RV  $Y = 1/X$ .

6.3. Let  $X$  be an RV with PDF

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let  $Y = \sin(2\pi X)$ . Find the DF and PDF of the RV  $Y$ .

6.4. Justify your answer or provide a counter-example:

- (a) Let  $X$  be a continuous RV. Does it follow that  $Y = |X|$  will be a continuous RV?
- (b) Let  $X$  be an RV and  $Y = |X|$  be a continuous RV. Does it imply that  $X$  is a continuous RV?

### Weeks 7 & 9 (8 Sep – 26 Sep)

7.1. Find the expected number of throws of a fair die until a 6 is obtained.

7.2. From a box containing  $N$  tickets numbered 1 through  $N$ ,  $n$  tickets are drawn with replacement. Let  $X$  be the largest number drawn. Find  $E[X]$ .

7.3. Pareto's distribution with parameters  $\alpha$  and  $\beta$  (both positive) is determined by the PDF

$$f(x) = \begin{cases} \frac{\beta \alpha^\beta}{x^{\beta+1}} & \text{if } x \geq \alpha, \\ 0 & \text{if } x < \alpha. \end{cases}$$

Show that the moment of order  $n$  exists if and only if  $n < \beta$ . If  $\beta > 2$ , find the mean and variance of the distribution.

7.4. For the Poisson RV  $X$  with PMF

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots,$$

compute  $E[X^n]$  and  $E[(X - \mu)^n]$  for  $n = 1, 2, 3, 4$ , where  $\mu = E[X]$ .

7.5. An RV  $X$  is called degenerate if there exists  $c \in \Omega$  for which  $P(X = c) = 1$ . Prove that  $\text{var}(X) = 0$  if and only if  $X$  is degenerate.

9.1. Let  $X$  be a non-negative RV,  $g : \mathbb{R} \rightarrow \mathbb{R}$  be a measurable function satisfying  $g(x) > 0$  and increasing for  $x > 0$ , and  $E[g(X)] < \infty$ . Show that for every  $a > 0$ ,

$$P(X > a) < \frac{E[g(X)]}{g(a)}.$$

9.2. Let  $X$  be an RV such that its MGF  $M(t) = E[e^{tX}]$  exists for every  $t > 0$ . Then prove that for any  $t > 0$ ,

$$P(tX > s^2 + \log M(t)) < e^{-s^2}.$$

9.3. For any RV  $X$ , show that

$$P(X \geq 0) \leq \inf[\varphi(t) : t \geq 0] \leq 1,$$

where  $\varphi(t) = E[e^{tX}]$ ,  $0 < \varphi(t) \leq \infty$ .

9.4. Let  $X$  be an RV such that  $P(X \in [a, b]) = 1$ , where  $-\infty < a < b < \infty$ . Show that

$$\text{var}(X) \leq \frac{(b-a)^2}{4}.$$

9.5. Use Chebyshev's inequality to show that for any  $k > 1$ ,  $e^{k+1} \geq k^2$ .

### Weeks 10 (29 Sep – 3 Oct)

10.1. A call center receives an average of 10 calls per minute. What's a good approximation to the probability that at most 3 calls are received in a given interval of 1 minute?

10.2. Let  $X$  be a memoryless RV. What can we say about the memorylessness of the RV  $X^2$ ?

10.3. Show that the terms

$$P_\lambda(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots,$$

reach their maxima when  $k$  is the largest integer  $\leq \lambda$ , and at  $\lambda - 1$  and  $\lambda$  if  $\lambda$  is an integer.

- 10.4. A biased coin is tossed indefinitely. Let  $p$  ( $0 < p < 1$ ) be the probability of success (heads). Let  $Y_1$  denote the length of run till first success, and  $Y_2$ , the length of the run from beginning to second success. Find the PMFs of  $Y_1$  and  $Y_2$ , and show that  $E[Y_1] = 1/p$ ,  $E[Y_2] = \frac{2}{p} - 1$ . If  $Y_n$  denotes the length till the  $n^{\text{th}}$  success for  $n \geq 1$ , what is the PMF of  $Y_n$ ? Find  $E[Y_n]$ .
- 10.5. A fair die is rolled  $n$  times. Find the number of trials needed so that the probability of at least one 6 to appear is at least  $1/2$ .

### Weeks 12 (12 Oct – 17 Oct)

- 12.1. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a multivariate RV on  $(\Omega, \mathcal{S})$  with  $X = (X_1, \dots, X_n)$  and  $X_i : \Omega \rightarrow \mathbb{R}$  for  $1 \leq i \leq n$ . Prove that each  $X_i$  is a univariate RV on  $(\Omega, \mathcal{S})$ .
- 12.2. Let  $\Omega = \{a, b, c, d, e\}$ ,  $\mathcal{S} = \mathcal{P}(\Omega)$  and  $\mathbb{P} : \mathcal{S} \rightarrow [0, 1]$  be given by  $\mathbb{P}(\{a\}) = 1/5$ ,  $\mathbb{P}(\{b\}) = 1/10$ ,  $\mathbb{P}(\{c\}) = 1/10$ ,  $\mathbb{P}(\{d\}) = 2/5$ ,  $\mathbb{P}(\{e\}) = 1/5$ . Let  $X$  be an RV on  $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$  given by  $X(a) = (1, 1)$ ,  $X(b) = (0, 1)$ ,  $X(c) = (0, 2)$ ,  $X(d) = (-1, 0)$ ,  $X(e) = (2, -1)$ . Plot the graph of the DF  $F(x, y)$  of  $X$ .
- 12.3. Let  $T$  be the right angled triangle with vertices  $(0, 0)$ ,  $(0, \sqrt{2})$ ,  $(\sqrt{2}, \sqrt{2})$ . Let  $F(x, y)$  be equal to the area of intersection of  $T$  with the region  $\mathcal{R}(x, y) = \{(a, b) : a \leq x, b \leq y\}$ . Show that  $F(x, y)$  is a DF on  $\mathbb{R}^2$  and find its marginal DFs.
- 12.4. Let  $(X, Y)$  have joint density function  $f$  and joint distribution function  $F$ . Suppose

$$f(x_1, y_1)f(x_2, y_2) \leq f(x_1, y_2)f(x_2, y_1),$$

holds for  $x_1 \leq a \leq x_2$  and  $y_1 \leq b \leq y_2$ . Show that

$$F(a, b) \leq F_1(a)F_2(b).$$

- 12.5. Let  $f(x, y, z) = e^{-x-y-z}$  for  $x > 0, y > 0, z > 0$ , and 0 otherwise be the joint PDF of  $(X, Y, Z)$ . Compute  $P(X < Y < Z)$  and  $P(X = Y < Z)$ .

- 12.6. For the DFs  $F, F_1, F_2, \dots, F_n$ , show that

$$1 - \sum_{i=1}^n [1 - F_i(x_i)] \leq F(x_1, \dots, x_n) \leq \min_{1 \leq i \leq n} F_i(x_i),$$

for all real  $x_1, \dots, x_n$  if and only if  $F_i$ 's are marginal DFs of  $F$ .

- 12.7. Is the function

$$f(x, y, z, u) = \begin{cases} e^{-u}, & 0 < x < y < z < u < \infty \\ 0, & \text{otherwise,} \end{cases}$$

a joint density function? If so, find  $\mathbb{P}(X \leq 7)$  where  $(X, Y, Z, U)$  is an RV with density  $f$ .

**Week 13 (19 Oct – 24 Oct)**

- 13.1. Suppose that the RV  $(X, Y)$  is uniformly distributed over the region  $R = \{(x, y) : 0 < x < y < 1\}$ . Find the covariance between  $X$  and  $Y$ .
- 13.2. Let  $(X, Y)$  be jointly distributed with PDF  $f$  defined by  $f(x, y) = 1/2$  inside the square with vertices  $(0, 1), (1, 0), (-1, 0), (0, -1)$ , and  $f(x, y) = 0$  otherwise. Are  $X$  and  $Y$  independent? Are they uncorrelated?
- 13.3. Let  $X_1, \dots, X_{m+n}$  be i.i.d. RVs with finite second moment. Let  $S_k = \sum_{i=1}^k X_i$  for  $k = 1, 2, \dots, m+n$ . Find the correlation coefficient between  $S_n$  and  $S_{m+n} - S_n$ .
- 13.4. Let  $\{X_n\}$  be a sequence of identically distributed RVs with mean  $\mu$ , variance  $\sigma^2$  and covariance (for  $i \neq j$ )  $\rho_{ij} = \text{cov}(X_i, X_j) \rightarrow 0$  as  $|i - j| \rightarrow \infty$ . Show that the Weak Law of Large Numbers holds for this sequence.
- 13.5. Let  $\{X_1, \dots, X_{10}\}$  be iid RVs uniformly distributed on  $(0, 1)$ . Use CLT to approximate

$$\mathbb{P}\left(\sum_{i=1}^{10} X_i > 6\right).$$

**Week 14 (26 Oct – 31 Oct)**

- 14.1. For a positive RV  $X$  with a finite first moment, show that  $E[\sqrt{X}] \leq \sqrt{E[X]}$  and  $E[1/X] \geq 1/E[X]$ .
- 14.2. Let  $X$  and  $Y$  be dependent RVs with means 0, variances 1 and correlation coefficient  $\rho$ . Show that

$$E[\max(X^2, Y^2)] \leq 1 + \sqrt{1 - \rho^2}.$$

- 14.3. If  $X$  is a nondegenerate RV with finite first moment, such that  $X \geq a > 0$ , show that

$$E[\sqrt{X^2 - a^2}] \leq \sqrt{E[X]^2 - a^2}.$$

- 14.4. Derive Lyapunov's inequality

$$E[|X|^r]^{1/r} \leq E[|X|^s]^{1/s}, \quad 1 < r < s < \infty,$$

from Hölder's inequality.

- 14.5. Let  $X_1, X_2, \dots$  be a sequence of i.i.d RVs with common PDF  $f(x) = e^{-x+\theta}$  if  $x \geq \theta$  and 0 otherwise. Write  $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ .

(a) Show that  $\bar{X}_n \xrightarrow{P} 1 + \theta$ .

(b) Show that  $\min\{X_1, \dots, X_n\} \xrightarrow{P} \theta$ .

- 14.6. Let  $X_1, X_2, \dots$  be iid  $U[0, \theta]$  RVs. Show that  $\max\{X_1, \dots, X_n\} \xrightarrow{P} \theta$ .